Introduction to Programming (II) Midterm Exam **Name:** **Student ID:**

1. How many lines will the program print? (2%) What is the output of the program? (6%) Explain why the variable *j* cannot be initialized in line 8. (2%)

```
1. #include <iostream>
2. using namespace std;
3. int main() {
4.     for (int i = 9; i <= 13; i++) {
5.         if (i % 2 == 0 || i < 11)
6.             continue;
7.         int j = 8;
8.         for (; j >= 5; j--) {
9.             if (j < 7)
10.                break;
11.            cout << (j^i) << endl;
12.        }
13.        cout << (double)(i++/j) << endl;
14.    }
15. }
```

2. Please describe the advantage and disadvantage of **inline function**. (5%) What kind of functions are suitable to be written as **inline functions**? (5%)
3. Write a simple C++ program to demonstrate "**function overloading**". You should also explain your program. (10%)
4. Write a simple C++ program to demonstrate "**function template**". You should also explain your program. (10%)
5. What is the output of the following program? (5%) What problem does the following program have? You should explain the problem in detail. (5%)

```
#include <iostream>
using namespace std;
int main() {
    int p = 6;
    int *q = new int;
    int *r = new int;
    *q = p;
    *r = 8;
    r = q;
    *r += (*q + p);
    cout << *q << endl;
    cout << *r << endl;
}
```

6. What is the output of the program below?

```
#include <iostream>
using namespace std;
int f(int a) {
    if (a <= 5)
        return 12;
    cout << a << endl;
    return f(a-2)*f(a-1);
}
int main() {
    cout << f(8) << endl;
}
```

7. Explain what the difference between the **smart pointer** and the **raw pointer** is. (10%)
8. Write three programs to demonstrate **Call by Value**, **Call by Address**, and **Call by**

Reference in C++. You should explain the programs you write. (10%)

9. What is the output of the program below? (10%)

```
#include <iostream>
using namespace std;
int * f(int c[3]) {
    c[2] = 8;
    return c;
}
int main() {
    int a[3] = {5, 2, 6};
    int *b;
    b = f(a);
    b[1] = a[0]+a[1];
    a[0] = *(b+2);
    for (int x: a)
        cout << x << endl;
}
```

10. (1) Figure out the lines that will cause compilation error. (5%) (2) After <u>removing the lines that will cause compilation error</u>, what is the output of this program? (5%)

```
1.  #include<iostream>
2.  using namespace std;
3.  class Midterm {
4.  private:
5.      int i1;
6.      static int si1;
7.  public:
8.      int i2;
9.      Midterm(int v) {
10.         i1 = v+1;
11.         i2 = v+3;
12.         si1 = v+5;
13.     }
14.     static int si2;
15.     static void setVariables(int x) {
16.         i1 = x+1;
17.         si2 = x+2;
18.     }
19. };
20.
21. int Midterm::si1 = 2;
22. int Midterm::si2 = 3;
23.
24. int main(int argc, const char * argv[]) {
25.
26.    Midterm midterm1(2);
27.    Midterm midterm2(3);
28.    midterm1.setVariables(5);
29.
30.    cout << Midterm::si1 << endl;
31.    cout << Midterm::si2 << endl;
32.
33.    midterm1.si2 = 5;
34.
35.    cout << midterm1.i1 << endl;
36.    cout << midterm2.i1 << endl;
37.    cout << midterm1.i2 << endl;
38.    cout << midterm2.i2 << endl;
39.    cout << midterm1.si1 << endl;
40.    cout << midterm2.si1 << endl;
41.    cout << midterm1.si2 << endl;
42.    cout << midterm2.si2 << endl;
43. }
```